

RUDRA

Radiometric Dynamic-Range Conditioning for HDR-Aware Diffusion Models

FXTD Studios / Radiance Research

Abstract

Recent diffusion models have achieved strong performance in image synthesis, restoration, and inpainting, but most pipelines remain centered on low dynamic range SDR image representations. This design compresses scene-referred luminance, highlight structure, exposure context, and wide-gamut color before the generative model can reason about them. This limitation is especially problematic for visual effects, cinematic post-production, HDR mastering, and OpenEXR pipelines, where scene-linear radiometric values are part of the production asset rather than a display artifact. We introduce RUDRA, a radiometric dynamic-range conditioning framework that augments a frozen latent diffusion backbone with a lightweight Dynamic Range Encoder and cross-attention based radiometric token injection. RUDRA converts SDR, HDR10, PQ, HLG, and OpenEXR inputs into scene-linear radiance, extracts a five-channel descriptor $R(x) = [L(x), E(x), H(x), x_CIE(x), y_CIE(x)]$, encodes it into a unified latent tensor Z_R , and injects this information into the diffusion backbone together with text conditioning. The revised manuscript clarifies the HDR reconstruction decoder, dataset protocol, descriptor dimensionality, ablation requirements, conditioning strength sweep, and qualitative evaluation protocol. Preliminary internal validation indicates improved highlight reconstruction, HDR perceptual quality, and exposure consistency over SDR-centric baselines; final public release should include the reproducibility package and qualitative sample grid described in this paper.

Keywords: diffusion models, HDR imaging, OpenEXR, radiometric conditioning, dynamic range, VFX, highlight reconstruction, latent diffusion, HDR-VDP-3

1. Introduction

Diffusion-based generative models have become the dominant approach for photorealistic image synthesis and restoration. Models such as DDPM, DDIM, latent diffusion models, SDXL, ControlNet, and newer flow-matching systems show high visual quality under common consumer imaging conditions. However, these models generally process normalized SDR tensors. The normalization step is not neutral for professional imaging: it collapses scene-referred radiance, clips or compresses highlights, and maps wide-gamut color volumes into a narrower display-referred RGB envelope.

Professional VFX and high-end cinematic workflows commonly use OpenEXR and HDR mastering formats because they preserve scene-linear values, exposure headroom, and compositing-critical highlight information. A diffusion system that first tone maps these assets to SDR discards information before modeling begins. RUDRA treats dynamic range as a first-class conditioning signal, analogous in importance to text, depth, segmentation, or pose conditioning, but targeted at radiometric structure rather than spatial layout.

The central hypothesis is that a frozen diffusion backbone can be made HDR-aware through a compact adapter that represents luminance, exposure, highlight energy, and chromaticity in a shared latent space. This allows the model to preserve radiometric intent without retraining the entire generator from scratch.

Contributions

- A corrected five-channel dynamic-range descriptor $R(x) = [L(x), E(x), H(x), x_CIE(x), y_CIE(x)]$ that separates luminance, exposure, highlight energy, and chromaticity.
- A Dynamic Range Encoder (DRE), implemented as a 12-layer transformer that maps the descriptor into a spatial latent tensor Z_R compatible with LDM-family backbones.

- A radiometric conditioning pathway that injects dynamic-range tokens through cross-attention, instead of treating HDR information as a simple RGB pre-processing step.
- A clarified HDR Reconstruction Decoder that outputs scene-linear HDR RGB suitable for OpenEXR export.
- A revised experimental protocol including shot-level dataset splitting, descriptor ablations, conditioning strength sweep, qualitative comparison grids, highlight crops, and video exposure consistency evaluation.

2. Related Work

Latent diffusion models reduce the cost of high-resolution synthesis by operating in a learned latent space and using cross-attention for conditioning. SDXL extends this design with a larger U-Net backbone and additional text-conditioning capacity. Conditional adapter systems such as ControlNet, T2I-Adapter, and IP-Adapter demonstrate that pretrained generators can be steered through auxiliary conditions. RUDRA follows the adapter philosophy but changes the conditioning target: it models scene radiometry instead of edges, depth, segmentation, pose, or image identity.

HDR reconstruction research includes multi-exposure fusion, single-image inverse tone mapping, and CNN or transformer based HDR expansion. These methods can reconstruct HDR-like outputs from SDR observations, but they are usually not designed as conditioning modules inside diffusion backbones. RUDRA occupies the intersection between HDR reconstruction and conditional diffusion: it uses HDR descriptors to steer a generative model while preserving compatibility with existing pretrained diffusion systems.

HDR assessment requires metrics that understand luminance adaptation and perceptual visibility in high dynamic range conditions. The revised evaluation therefore includes HDR-VDP-3, Delta E 2000, Highlight Reconstruction Accuracy, EV error, LPIPS, PSNR, and SSIM. SDR metrics are retained only as secondary indicators after controlled tone mapping.

3. Method

3.1 Overview

RUDRA augments a frozen LDM-family diffusion backbone with three trainable components: a format normalization and Dynamic Range Analysis front end, a Dynamic Range Encoder, and a radiometric cross-attention conditioning pathway. The decoder stage is HDR-aware and returns scene-linear RGB rather than display-referred SDR.

Figure 1 shows the revised architecture. The figure incorporates the main review corrections: format normalization is explicit, the descriptor is five-dimensional, and dynamic-range conditioning is represented as token injection into cross-attention rather than plain scalar addition.

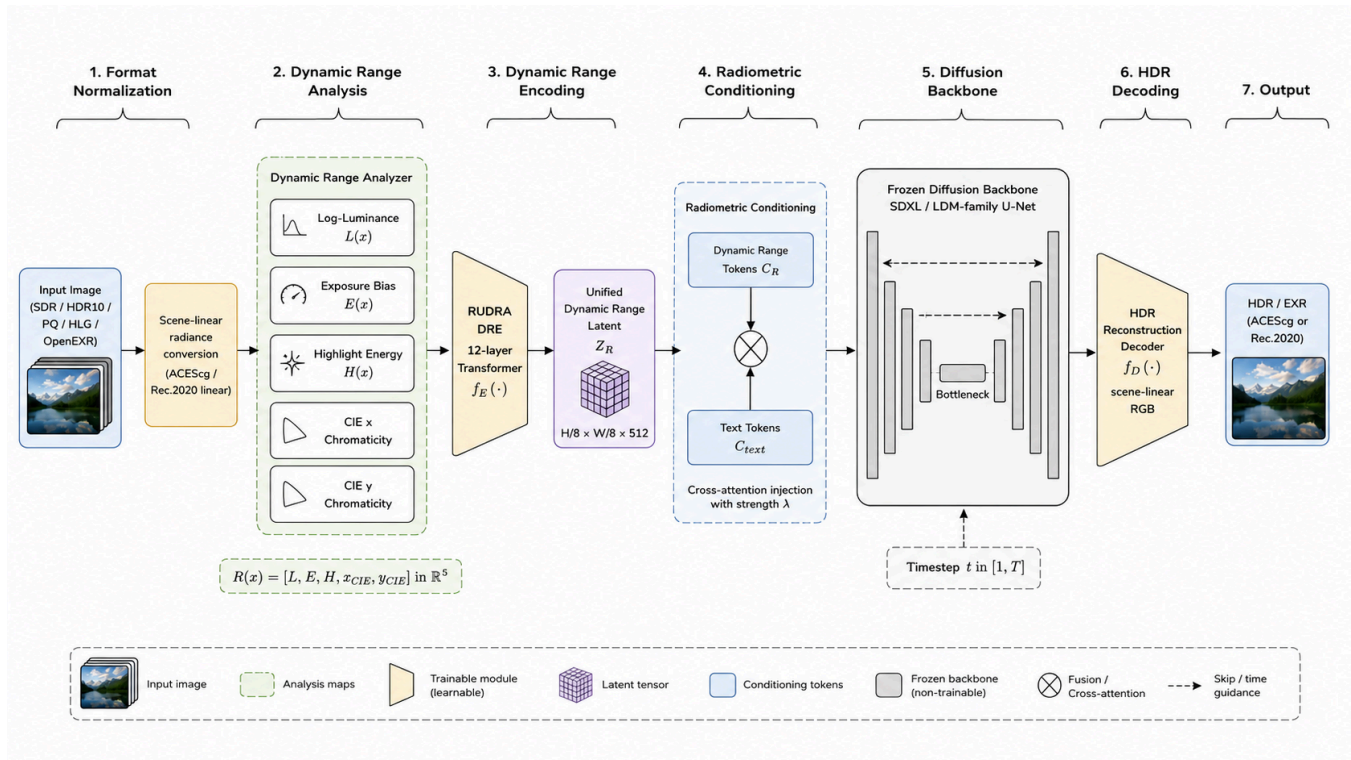


Figure 1. Revised RUDRA architecture. The pipeline converts heterogeneous formats to scene-linear radiance, extracts a five-channel radiometric descriptor, encodes it into Z_R , injects radiometric tokens into a frozen diffusion backbone, and decodes scene-linear HDR/OpenEXR output.

3.2 Format Normalization

All inputs are converted to scene-linear radiance before descriptor extraction. SDR PNG and JPEG images are linearized from display-referred RGB using the known transfer function or an estimated inverse gamma when metadata is unavailable. HDR10 and PQ inputs are inverted with the SMPTE ST 2084 electro-optical transfer function to recover absolute luminance. HLG inputs are inverted using the HLG opto-electrical transfer function and mapped into the selected scene-linear working space. OpenEXR inputs are treated as scene-linear by default, with color space tags used when available. The reference implementation should select one output working space for training and evaluation, preferably ACEScg for VFX workflows or Rec.2020 linear for HDR video workflows.

3.3 Corrected Dynamic-Range Descriptor

For every pixel x in the scene-linear image I , RUDRA computes a five-channel descriptor:

$$R(x) = [L(x), E(x), H(x), x_{CIE}(x), y_{CIE}(x)] \text{ in } \mathbb{R}^5$$

This corrects the earlier inconsistency where CIE xy chromaticity was described as a two-component quantity but the descriptor was still labeled $\mathbb{R}^4$. The descriptor channels are:

Channel	Definition	Purpose
$L(x)$	$\log_{10}(1 + Y(x)) / \log_{10}(1 + Y_{\max})$	Compresses luminance while preserving relative HDR ordering.
$E(x)$	$\log_2(\text{median}(Y) / 0.18)$, broadcast spatially	Encodes global exposure bias in EV stops.
$H(x)$	$\max(0, L(x) - \tau) / (1 - \tau)$	Provides a smooth highlight energy mask.
$x_{CIE}(x), y_{CIE}(x)$	CIE 1931 chromaticity from scene-linear RGB converted to XYZ	Separates hue/chromaticity from luminance.

3.4 Dynamic Range Encoder

The Dynamic Range Encoder maps R in $R^{(H \times W \times 5)}$ to a compact spatial latent Z_R . The descriptor map is patchified into 8×8 non-overlapping patches. Each patch is projected into a 512-dimensional token embedding with learned positional encodings. A 12-layer transformer encoder with 8-head self-attention, pre-layer normalization, GELU activations, and MLP expansion factor 4 processes the token grid. The output is reshaped to Z_R in $R^{(H/8 \times W/8 \times 512)}$, matching the spatial resolution expected by common latent diffusion backbones.

This module is the primary trainable radiometric representation learner. In the reference design, the diffusion backbone, text encoder, and base VAE encoder remain frozen, while DRE and conditioning projection layers are trained.

3.5 Radiometric Conditioning

Instead of adding a single pooled vector to text embeddings, RUDRA should inject radiometric tokens into selected U-Net cross-attention blocks. Let C_{text} denote the text-conditioning tokens and C_R denote the projected dynamic-range tokens derived from Z_R . The conditioning set is:

$$C_{\text{total}} = \text{concat}(C_{\text{text}}, \lambda * C_R)$$

where λ controls radiometric conditioning strength. This representation is more faithful than $C_{\text{text}} + \lambda C_R$ because it preserves token identity and allows attention heads to separately attend to semantic and radiometric sources. The reference value $\lambda = 0.8$ is retained as the preliminary operating point, but the revised evaluation requires a conditioning strength sweep.

3.6 HDR Reconstruction Decoder

The HDR Reconstruction Decoder f_D maps the denoised latent into scene-linear HDR RGB. This component must be explicitly defined because HDR output cannot be treated as a normal SDR VAE decode. The revised reference decoder uses a modified VAE decoder head with a non-clamping output parameterization:

- Output space: scene-linear RGB in ACEScg or Rec.2020 linear, selected once for the experiment.
- Numerical range: values are allowed above 1.0 to preserve HDR highlights; optional softplus or signed log-radiance parameterization may be used to avoid hard clipping.
- Precision: training uses float16 or bfloat16 activations, but export writes half-float or float OpenEXR.
- Color management: all evaluation is performed after converting predictions and ground truth into the same working space.
- Negative values: small negative reconstruction values are either preserved for ACEScg/VFX compatibility or clipped only at export, but the rule must be fixed before evaluation.
- Export: final HDR outputs are written as OpenEXR with explicit chromaticities, transfer metadata, and optional luminance scale metadata.

This section is essential for reproducibility because an HDR-aware conditioning branch is incomplete if the final decoder clamps results back into SDR.

4. Training Objective

The training objective combines diffusion denoising, highlight preservation, chromaticity preservation, perceptual structure, and optional latent alignment:

$$L = L_{\text{diff}} + \alpha L_{\text{highlight}} + \beta L_{\text{color}} + \gamma L_{\text{perceptual}} + \eta L_{\text{align}}$$

The preliminary weights are $\alpha = 0.50$, $\beta = 0.30$, $\gamma = 0.20$. The revised paper should report the selected η for HDR latent alignment and include a validation sweep for all major weights. Highlight loss is masked by $H(x)$ and computed in scene-linear luminance. Color loss is computed in CIE xy or a perceptually uniform color space after

fixing the working color space. LPIPS is computed on consistently tone-mapped SDR previews only, because standard LPIPS backbones are not HDR-native.

5. Experimental Protocol

5.1 Dataset and Split Policy

The preliminary dataset contains 17,100 training pairs and 3,420 validation pairs across HDR captures, OpenEXR renders, VFX shots, synthetic physically based renders, and calibrated SDR/HDR pairs. The revised protocol requires that train and validation splits are performed by scene, shot, or environment, not by individual frame. This prevents temporal leakage from near-duplicate video frames or panoramas.

Source	Train	Validation	Formats	Split requirement
HDR-Eye / HDR survey	2,000	400	EXR, HDR10	Split by capture scene
Laval Indoor HDR	2,100	420	OpenEXR / HDR panorama	Split by environment
OpenEXR Test Suite	800	160	OpenEXR	Split by render asset
VFX Shot Library	3,200	640	EXR, PQ	Split by shot, not frame
Synthetic HDR Scenes	4,000	800	EXR, HLG	Split by scene seed and camera setup
Calibrated SDR Pairs	5,000	1,000	PNG/JPEG + HDR	Split by source sequence
Total	17,100	3,420	Mixed	No near-duplicate leakage

5.2 Baselines

Baselines should include a vanilla latent diffusion pipeline with tone-mapped inputs, SDXL without RUDRA conditioning, a flow-matching or transformer-based text-to-image model evaluated through the same HDR proxy pipeline, a single-image HDR reconstruction model, and an inverse tone-mapping network. All baselines must receive the same input resolution, same tone-mapping policy when required, and the same evaluation color space. If a baseline is SDR-only, the paper must state exactly how its output is converted to HDR for comparison.

5.3 Metrics

Metric	Domain	Purpose
HDR-VDP-3	HDR luminance	Primary perceptual HDR quality score
HRA	Highlight mask $H(x) > 0$	Measures highlight reconstruction within relative luminance tolerance
Delta E 2000	Color difference	Measures color fidelity after common color-space conversion
EV error	Frame/shot exposure	Measures exposure consistency in video or sequence data
LPIPS	Tone-mapped SDR preview	Secondary perceptual structure measure
PSNR / SSIM	Tone-mapped SDR preview	Secondary reconstruction fidelity metrics

6. Preliminary Results

The following results are retained from the current internal draft as preliminary validation. They should be presented as preliminary until the reproducibility package, qualitative grids, and complete ablation suite are released.

Method	PSNR (dB) up	SSIM up	LPIPS down	HDR-VDP-3 up	HRA (%) up
SDR Latent Diffusion	27.8	0.795	0.201	65.4	58.9
Stable Diffusion XL	28.4	0.812	0.189	67.2	61.3
FLUX.1 SDR proxy	29.1	0.831	0.174	68.9	63.7
Single-frame HDR-CNN	30.2	0.847	0.162	70.1	67.4
Inverse Tone Mapping	31.0	0.855	0.155	71.8	68.9
RUDRA preliminary	33.7	0.891	0.121	80.4	78.9

Interpretation: RUDRA shows the largest gain in HDR-VDP-3 and HRA, which is consistent with the claim that radiometric conditioning mainly helps highlight and luminance reconstruction. However, final claims should remain conservative until the qualitative and reproducibility requirements are met.

7. Required Ablation Suite

7.1 Component Ablation

Configuration	HDR-VDP-3 up	LPIPS down	PSNR up	HRA up
Baseline diffusion	67.2	0.189	28.4	61.3
+ Dynamic Range Encoder	71.5	0.164	29.9	65.1
+ HDR latent alignment	74.8	0.148	31.2	69.4
+ Highlight preservation loss	76.9	0.135	32.1	73.7
+ HDR color preservation loss	78.6	0.128	32.8	76.2
Full RUDRA	80.4	0.121	33.7	78.9

7.2 Descriptor Channel Ablation

The revised paper must include a channel-level ablation to prove that each part of $R(x)$ is necessary. The table below is a publication-ready template. Fill with measured values after running the experiment.

Descriptor variant	Expected purpose	HDR-VDP-3	Delta E 2000	HRA	EV error
L only	Tests luminance-only radiometry	TBD	TBD	TBD	TBD
L + E	Tests exposure conditioning	TBD	TBD	TBD	TBD
L + H	Tests highlight mask contribution	TBD	TBD	TBD	TBD
L + x_CIE + y_CIE	Tests chromaticity without exposure/highlight	TBD	TBD	TBD	TBD
Full R^5 descriptor	Complete RUDRA descriptor	TBD	TBD	TBD	TBD

7.3 Conditioning Strength Sweep

The paper should report performance across lambda values to show the stability of radiometric guidance. This prevents the method from appearing tuned to one unpublished hyperparameter.

lambda	HDR-VDP-3	LPIPS	HRA	Delta E 2000	Observed behavior
0.00	TBD	TBD	TBD	TBD	No radiometric conditioning
0.25	TBD	TBD	TBD	TBD	Weak radiometric guidance
0.50	TBD	TBD	TBD	TBD	Moderate guidance
0.80	TBD	TBD	TBD	TBD	Preliminary selected value
1.00	TBD	TBD	TBD	TBD	Strong guidance
1.25	TBD	TBD	TBD	TBD	Check over-conditioning artifacts

8. Qualitative Evaluation Plan

A final arXiv release should include at least one full-page qualitative grid. Each row should show the input, SDXL baseline, HDR-CNN or inverse tone mapping baseline, RUDRA prediction, ground truth HDR, and a highlight error heatmap. Crops should focus on specular metal, neon lights, fire, sun reflections, windows, headlights, and bright sky gradients.

For video or frame sequences, the paper should include an EV-error plot over time. This validates the claim that exposure conditioning reduces frame-to-frame drift in VFX shots.

9. Reproducibility Checklist

- Publish exact preprocessing scripts for SDR, PQ, HLG, and OpenEXR normalization.
- State the training working color space and the export color space.
- Publish train/validation split files by scene or shot.
- Release model configuration for DRE depth, token dimension, attention heads, and projection layers.
- Report optimizer, learning-rate schedule, batch size, training steps, precision mode, and GPU count.
- Define baseline conversion rules, especially for SDR-only methods evaluated against HDR ground truth.
- Release qualitative comparison samples or a reproducible evaluation notebook.
- Mark private/proprietary datasets clearly and provide public substitutes when licensing prevents release.

10. Limitations

RUDRA increases memory use through the Dynamic Range Encoder and adds complexity to data handling because all inputs must be normalized into a consistent scene-linear space. The current validation corpus may underrepresent sparse night scenes with isolated light sources, extreme laser or LED colors, and shots with strong lens artifacts. Temporal consistency is only partially addressed by exposure conditioning; a video-native RUDRA variant should add temporal attention or temporal losses. Porting from SDXL to flow-matching or transformer-based image generators may require architecture-specific projection layers. Finally, the preliminary numerical results should be treated as internal validation until the public reproducibility package is available.

11. Conclusion

RUDRA reframes HDR-aware diffusion as a radiometric conditioning problem. By extracting a corrected five-channel descriptor and encoding it into a unified dynamic-range latent space, the method gives a frozen diffusion backbone explicit access to luminance, exposure, highlight structure, and chromaticity. The revised manuscript strengthens the original concept by clarifying the decoder, correcting descriptor dimensionality, defining dataset leakage controls, and specifying the missing ablation and qualitative evidence required for a credible arXiv release. With the additional experiments completed, RUDRA has strong potential as a practical path toward HDR-native generative pipelines for VFX, cinematic post-production, and OpenEXR-based computational imaging.

References

1. Ho, J., Jain, A., and Abbeel, P. Denoising Diffusion Probabilistic Models. NeurIPS, 2020.
2. Song, J., Meng, C., and Ermon, S. Denoising Diffusion Implicit Models. ICLR, 2021.
3. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-Resolution Image Synthesis with Latent Diffusion Models. CVPR, 2022. arXiv:2112.10752.
4. Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Mueller, J., Penna, J., and Rombach, R. SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. arXiv:2307.01952, 2023.
5. Zhang, L., Rao, A., and Agrawala, M. Adding Conditional Control to Text-to-Image Diffusion Models. ICCV, 2023.

6. Mou, C., Wang, X., Xie, L., Wu, Y., Zhang, J., Qi, Z., and Shan, Y. T2I-Adapter: Learning Adapters to Dig out More Controllable Ability for Text-to-Image Diffusion Models. AAAI, 2024.
7. Ye, H., Zhang, J., Liu, S., Han, X., and Yang, W. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. arXiv:2308.06721, 2023.
8. Debevec, P. and Malik, J. Recovering High Dynamic Range Radiance Maps from Photographs. SIGGRAPH, 1997.
9. Eilertsen, G., Kronander, J., Denes, G., Mantiuk, R., and Unger, J. HDR Image Reconstruction from a Single Exposure Using Deep CNNs. ACM Transactions on Graphics, 2017.
10. Liu, Y.-L., Lai, W.-S., Chen, Y.-S., Kao, Y.-L., Yang, M.-H., Chuang, Y.-Y., and Huang, J.-B. Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline. CVPR, 2020.
11. Luo, M. R., Cui, G., and Rigg, B. The Development of the CIE 2000 Colour-Difference Formula: CIEDE2000. Color Research and Application, 2001.
12. Mantiuk, R. K., Hammou, D., and Hanji, P. HDR-VDP-3: A Multi-Metric for Predicting Image Differences, Quality and Contrast Distortions in High Dynamic Range and Regular Content. arXiv:2304.13625, 2023.
13. Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR, 2021.
14. Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. CVPR, 2018.
15. Reinhard, E., Stark, M., Shirley, P., and Ferwerda, J. Photographic Tone Reproduction for Digital Images. ACM Transactions on Graphics, 2002.
16. Kainz, F., Bogart, R., and Hess, D. The OpenEXR Image File Format. ACM SIGGRAPH Technical Sketches, 2004.